

E-commerce Checkout & Cart Abandonment Improvement

Retail / E-commerce • Business Analyst Portfolio Project

Document Type	Business Requirements Document (BRD)
Project Type	Simulated Business Case using a public Kaggle dataset
Role	Business Analyst
Primary Tools	Excel • Power BI • Draw.io • Jira • Word/PDF
Data Period	01 Jan 2024 – 30 Dec 2024
Dataset Size	25,000 sessions • 8,442 unique customers
Version	1.0

Confidentiality / Source Note: This is a portfolio project based on a publicly available Kaggle e-commerce dataset. Stakeholder interviews and business requirements are simulated for demonstration of Business Analysis methods and are not represented as confidential information from a real company.

1. Executive Summary

The purpose of this project is to analyze the e-commerce cart and checkout journey, identify where customers abandon their carts, investigate the business factors associated with abandonment, and define a redesigned checkout experience. The project demonstrates an end-to-end Business Analyst approach: stakeholder discovery, requirements gathering, process analysis, data analysis, root-cause analysis, solution definition, user stories, acceptance criteria, KPI design, and dashboarding.

Baseline from the uploaded dataset: 16,117 sessions added an item to cart; 10,501 were marked as abandoned and 5,616 resulted in purchase. This gives a cart abandonment rate of 65.15% and a purchase conversion rate of 34.85% among cart-add sessions. Recorded revenue is ₹10,116,169.06.

These figures are baseline observations, not proof of causation. The subsequent analysis will test differences across device type, user type, marketing channel, product category, payment method, discount, cart value, and other available dimensions.

2. Business Problem

The e-commerce business is experiencing substantial cart abandonment. Customers add products to their carts but a significant proportion do not complete the purchase. This creates a gap between customer intent and realized revenue. The business needs a data-supported understanding of the checkout journey and a practical set of requirements for reducing friction.

Problem Statement

High cart abandonment is reducing completed orders and potential revenue. The business wants to identify the main drop-off patterns and define an improved checkout flow that is simpler, clearer, and easier to complete.

Business Objectives

- Measure the current cart abandonment and purchase conversion baseline.
- Identify high-drop-off customer segments and checkout-related patterns.
- Identify potential root causes using quantitative analysis and stakeholder input.
- Define functional and non-functional requirements for an improved checkout experience.
- Design an As-Is and To-Be checkout process.
- Create measurable KPIs to evaluate the redesigned checkout flow.

3. Scope

In Scope

- Cart and checkout customer journey analysis.
- Analysis of cart-add, purchase, and abandonment behavior.
- Segmentation by available customer/session dimensions.
- Requirements for checkout experience improvements.
- As-Is and To-Be process mapping.
- User stories and acceptance criteria.
- KPI definition and Power BI dashboard.

Out of Scope

- Actual production-system development or deployment.
- Real customer interviews or collection of personal customer information.
- Changes to payment gateway infrastructure.

- Live A/B testing of the proposed checkout flow.
- Financial forecasting beyond the dataset's recorded revenue.

4. Stakeholder Analysis

Stakeholder	Role / Interest	BA Need
E-commerce Manager	Business performance, sales, abandonment	Business goals and success criteria
Product Manager	Checkout product experience	Functional priorities and product constraints
UX/UI Designer	Usability and screen flow	Design requirements and user friction points
Payment Team	Payment processing	Payment failure and retry requirements
Customer Support	Customer complaints and support volume	Common checkout pain points
Data Analyst	Data quality and reporting	Metrics, dimensions and validation
Customers	End users completing purchases	Usability and friction requirements

5. Stakeholder Discovery - Simulated Interviews

Because this is a portfolio case, stakeholder conversations are simulated. The questions below represent a realistic discovery approach a BA could use before finalizing requirements.

Stakeholder	Question	Simulated Response / Requirement Signal
E-commerce Manager	What is the main business concern?	High cart abandonment and lost completed orders.
E-commerce Manager	What outcome should improve?	More customers completing checkout and less abandonment.
Product Manager	What checkout issues should be investigated?	Long or complex flow, mandatory account steps, unclear costs and payment friction.
Customer Support	What customer concerns are common?	Questions around delivery charges, payment issues and checkout difficulty.
Payment Team	What should happen after payment failure?	Customer should see a clear message and be able to retry without restarting the entire process.
UX Designer	What experience principles matter?	Fewer steps, clear information, mobile usability and visible progress.

6. Business Requirements

ID	Requirement	Priority	Success Measure
BR-01	Reduce cart abandonment through a simpler checkout journey.	High	Lower abandonment rate
BR-02	Increase the proportion of cart-add sessions that complete purchase.	High	Higher purchase conversion
BR-03	Make final payable cost clear before payment.	High	Lower cost-related drop-off
BR-04	Support a low-friction checkout option for customers who do not want to register first.	High	Higher checkout completion
BR-05	Reduce payment-related friction and provide payment retry.	High	Lower payment failure / recovery rate
BR-06	Measure checkout performance by customer and session segments.	Medium	Dashboard coverage

7. Data Requirements & Dataset Understanding

The analysis uses the uploaded CSV dataset supplied for this project. The dataset contains 29 fields covering customer/session identifiers, visit information, device and user type, marketing channel, product information, pricing/discounts, engagement, cart/purchase outcomes, payment method, and location.

Category	Relevant Fields
Identifiers	customer_id, session_id
Date / time	visit_date, visit_day, visit_month, visit_weekday, visit_season
Customer / session	device_type, user_type, location, session_duration_bucket
Product / order	product_id, product_category, unit_price, quantity
Commercial	discount_percent, discount_amount, revenue, revenue_normalized
Engagement	pages_viewed, time_on_site_sec
Conversion	added_to_cart, purchased, cart_abandoned
Payment / feedback	payment_method, rating, review_text, review_helpful_votes

Data Quality Baseline

Check	Result
Rows	25,000
Columns	29
Unique customers	8,442
Unique sessions	25,000
Duplicate rows	0
Missing cells	0
Date range	01 Jan 2024 - 30 Dec 2024

8. Baseline Business Metrics

Metric	Baseline
Total sessions	25,000
Cart-add sessions	16,117
Purchased sessions	5,616
Abandoned cart sessions	10,501
Cart abandonment rate	65.15%
Purchase conversion among cart-add sessions	34.85%
Recorded revenue	₹10,116,169.06

Calculation: Cart abandonment rate = abandoned cart sessions ÷ cart-add sessions × 100. Purchase conversion = purchased sessions ÷ cart-add sessions × 100.

Analytical caution: The dataset is session-level and publicly sourced. Therefore, these metrics should be used as portfolio baseline indicators rather than claims about a real company's current performance.

9. Analytical Questions

- At which customer/session segments is abandonment highest?
- How does abandonment vary by device type?

- How does behavior differ between new and returning users?
- Are some marketing channels associated with higher abandonment?
- Are particular product categories associated with higher drop-off?
- Does payment method show meaningful differences in purchase completion?
- How do discount levels and cart value relate to purchase completion?
- Does engagement (pages viewed or time on site) differ between purchasers and abandoners?
- Which findings are strong enough to become product or process requirements?

10. As-Is Checkout Process

The current-state process is a portfolio-level process hypothesis. It should be validated against available data and, in a real project, confirmed through stakeholder interviews and product analytics.

Step	Current Activity	Potential BA Question
1	Customer browses products	Is product intent strong enough to proceed?
2	Customer adds product to cart	What proportion continues?
3	Customer reviews cart	Are price, discount and quantity clear?
4	Customer enters account / customer details	Does registration create friction?
5	Customer enters address	Is address entry simple?
6	Customer selects delivery	Are delivery charges clear?
7	Customer reviews order	Is final cost visible?
8	Customer selects payment	Are preferred methods available?
9	Payment processed	What happens after failure?
10	Order confirmation	Is confirmation immediate and clear?

11. Functional Requirements

ID	Functional Requirement	Priority
FR-01	The checkout should provide a guest checkout option.	High
FR-02	The customer should be able to review and edit cart items before payment.	High
FR-03	The system should display product price, discount, delivery charge and final payable amount before payment.	High
FR-04	The customer should be able to enter and edit the delivery address.	High
FR-05	The checkout should display available payment methods clearly.	High
FR-06	If payment fails, the system should show a clear error message.	High
FR-07	After a payment failure, the customer should be able to retry payment without re-entering all checkout information.	High
FR-08	After successful payment, the system should display an order confirmation.	High
FR-09	The checkout should display progress/status so the customer knows the current step.	Medium

12. Non-Functional Requirements

ID	Requirement
NFR-01	Usability: checkout should be simple and understandable for first-time users.

ID	Requirement
NFR-02	Responsiveness: checkout should work effectively across supported mobile and desktop screen sizes.
NFR-03	Performance: checkout screens should load and respond within the application's agreed performance target.
NFR-04	Security: payment and customer information must be handled according to applicable security requirements.
NFR-05	Availability: checkout should be available according to the platform's agreed service-level target.

13. To-Be Checkout Flow

The proposed future-state flow aims to reduce unnecessary friction while preserving required customer, delivery, payment and confirmation steps.

Step	To-Be Experience
1	Cart review with editable items and clear subtotal
2	Guest checkout or login
3	Customer details / delivery address
4	Delivery option and clearly displayed charge
5	Order summary with final payable amount
6	Payment method selection
7	Payment success → confirmation; payment failure → retry

14. User Stories & Acceptance Criteria

ID	User Story	Acceptance Criteria
US-01	As a customer, I want to checkout as a guest so that I can proceed with purchase	Guest checkout is available; mandatory details can be entered; customer can continue to payment without login
US-02	As a customer, I want to see the complete payable amount before payment	Product price, discount, delivery charge and final amount are displayed before payment.
US-03	As a customer, I want to retry a failed payment so that I can complete my purchase	Failure message is shown; retry option is available; cart/address details are retained.
US-04	As a customer, I want to edit my cart before payment so that I can adjust my order	Quantity and price can be changed; items can be removed; totals update correctly.
US-05	As a customer, I want to see checkout progress so that I can understand the process	Current step is visible; completed steps are indicated.

15. KPI Framework

KPI	Definition	Purpose
Cart Abandonment Rate	$\text{Abandoned cart sessions} \div \text{cart-add sessions} \times 100$	Measure drop-off
Purchase Conversion Rate	$\text{Purchased sessions} \div \text{cart-add sessions} \times 100$	Measure checkout completion
Payment Failure Rate	$\text{Failed payment attempts} \div \text{payment attempts} \times 100$	Monitor payment friction
Average Cart Value	$\text{Cart value} \div \text{cart-add sessions}$	Understand commercial value of carts
Revenue per Session	$\text{Revenue} \div \text{total sessions}$	Track revenue efficiency
Abandonment by Device	Abandonment rate segmented by device	Identify device-specific friction
Abandonment by User Type	Abandonment rate segmented by user type	Compare new/returning behavior

16. Power BI Dashboard Requirements

Section	Visual / Measure	Business Purpose
Executive KPIs	Sessions, Cart Adds, Abandonment Rate, Purchase Conversion Rate	Quick performance view
Conversion Funnel	Session → Cart → Purchase/Abandonment	Show journey drop-off
Device Analysis	Abandonment rate by device	Find device-specific patterns
User Analysis	Abandonment rate by user type	Compare customer groups
Marketing Analysis	Abandonment by marketing channel	Assess traffic quality
Product Analysis	Abandonment by product category	Identify category patterns
Payment Analysis	Purchase/abandonment by payment method	Identify payment patterns
Trend	Abandonment and conversion over time	Monitor changes

17. Root Cause Analysis Approach

The project should not assume that a correlation is automatically a root cause. Root-cause analysis will combine quantitative evidence from the dataset with the simulated stakeholder discovery.

- Step 1 – Identify the largest abandonment segments.
- Step 2 – Compare the segment against the overall baseline.
- Step 3 – Check related variables such as device, payment, user type, discount and engagement.
- Step 4 – Use 5 Whys / Fishbone analysis to structure possible causes.
- Step 5 – Validate whether the evidence supports a requirement.
- Step 6 – Translate validated findings into To-Be process changes and measurable KPIs.

18. Assumptions & Constraints

- Stakeholder interviews are simulated because this is a portfolio case.
- The dataset is public and may not represent the behavior of any specific real retailer.
- The project focuses on analytical and BA deliverables rather than software implementation.
- Coded categorical variables must be mapped using the dataset's documented definitions before final dashboard labels are published.
- Any recommendation should be treated as a hypothesis until validated through experimentation or real stakeholder/user evidence.

19. Traceability - Business Need to Solution

Business Need	Requirement	Solution	KPI
Reduce abandonment	FR-01, FR-09	Guest checkout + progress visibility	Cart Abandonment Rate
Improve price transparency	FR-03	Full cost before payment	Purchase Conversion Rate
Reduce payment friction	FR-05, FR-06, FR-07	Clear methods + retry flow	Payment Failure Rate
Improve cart control	FR-02	Editable cart	Purchase Conversion Rate
Measure impact	BR-06	Power BI KPI dashboard	All core KPIs

20. BA Deliverables & Tool Mapping

Deliverable	Tool	Output
BRD / Requirements	Word → PDF	Professional documentation
Data cleaning	Excel	Clean analysis-ready dataset
Data analysis & dashboard	Power BI	Interactive KPI dashboard
As-Is / To-Be process	Draw.io	Process maps
Backlog / user stories	Jira	Epics, stories, acceptance criteria
Wireframes	Not used	Solution flow documented without a separate UI design tool

21. Acceptance / Sign-Off Criteria for the Portfolio Case

- Business problem, objective and scope are clearly documented.
- Stakeholders and simulated requirements are traceable.
- Dataset quality checks are completed.
- Baseline KPIs are calculated consistently.
- As-Is and To-Be flows are documented.
- Functional requirements and user stories have acceptance criteria.
- Power BI dashboard contains the defined KPI set.
- Recommendations are linked to observed evidence and clearly separated from assumptions.

22. Final Note

This BRD is designed as a professional portfolio artifact for a fresher Business Analyst profile. The next implementation steps are to clean and analyze the dataset in Excel, build the analytical model and dashboard in Power BI, create As-Is/To-Be process diagrams in Draw.io, and manage the requirements backlog in Jira. The final portfolio should show the connection between business problem, evidence, requirements, process improvement and measurable outcomes.